Introduction

Gostnet is a Linux-based free and open source reticulum chat platform inspired by the work of S2Underground. The intention of this project is to provide any community with an easy to deploy, reasonably secure messaging service with transport across mesh radio and TCP networks. The creator of this program did not invent reticulum. This is merely a front end for other people's work and is intended to allow the "average joe" a means of communication for recreation, experimentation or in times of emergency.

Because reticulum is encrypted by default, mind your local laws when deploying across radio networks. This may be illegal depending on your area in the world.

For online installation using terminal:
-Move the .deb file to a preferred location on a Linux machine and install using apt
  Example: sudo apt install /path/meridian-ghostnet_1.3.2_amd64.deb

For offline installation using terminal:
-Move the .tar.gz file to a preferred location on a Linux machine
-Navigate to the location you moved the .tar.gz file.
  Example: cd /path/to/meridian-ghostnet_1.3.2_amd64.tar.gz
-Extract the .tar.gz file
  Example: tar xzf meridian-ghostnet_1.3.2_amd64.tar.gz
-Navigate to the newly created directory
  Example: cd /path/to/meridian-ghostnet_1.3.2_tarball
-Install
  Example: sudo bash install.sh

After installation, Ghostnet will be served on http://localhost:8090 in your web browser or across your local network at your station's ip address on port 8090 (XX.XX.XX.XXX:8090). A login screen for a server based deployment can be configured in settings once the program is launched.

The creator of Ghostnet relied HEAVILY on AI coding agents because he's not very good at coding. Therefore, he won't be of much help to you in troubleshooting problems with the source code. If you're smarter than the creator of Ghostnet, you should improve it and publish it to the community.

Basic Station Configuration

Upon deployment, the operator will be presnted with the main screen that consists of two main panes: Station Configuration and Contacts.

1. Station configuration: Configure your RNS identity and interfaces here.

By default a LAN interface will be deployed and other interfaces can be added. If "full router" is configured as a role, all of the configured interfaces will create a mesh network.

Interfaces include:
a. LAN: All stations on the same subnet and network group will form a mesh
b. TCP: Client: Connect your device to a know IP / FQDN and port of a distant network over TCP
c. RNode LoRa: Connect to a RNode flashed LoRa device.
d. KISS Modem: Connect to a radio. The following radio modems were used during lab testing:
    -DigiRig Mobile: RTS
    -DigiRig Lite: VOX
    -Yaesu SCU-17: DTR
e. RNS Propogation Node: Creats a RNS propogation node with its own RNS identity and allows connected stations to store encrypted messages for offline message retrieval as well as allows other networks to connect over the greater internet. This is intended to be deployed on reliable "always on" stations.

Routing roles are consitant with publically available information on Reticulum networks.

***RF Links have a limitation (announce_cap=10) that is hard coded to prevent heavy traffic and preserve radio duty cycle. This will limit station announcements on a busy TCP connected mesh and should be a planning consideration when building TCP->Radio mesh networks.

2. The contacts page has two sub tabs; Phone Book and Network

The Network tab is live and reflects all of the stations heard on your network. The operator may check the box next to a station to add them to the Phone Book.

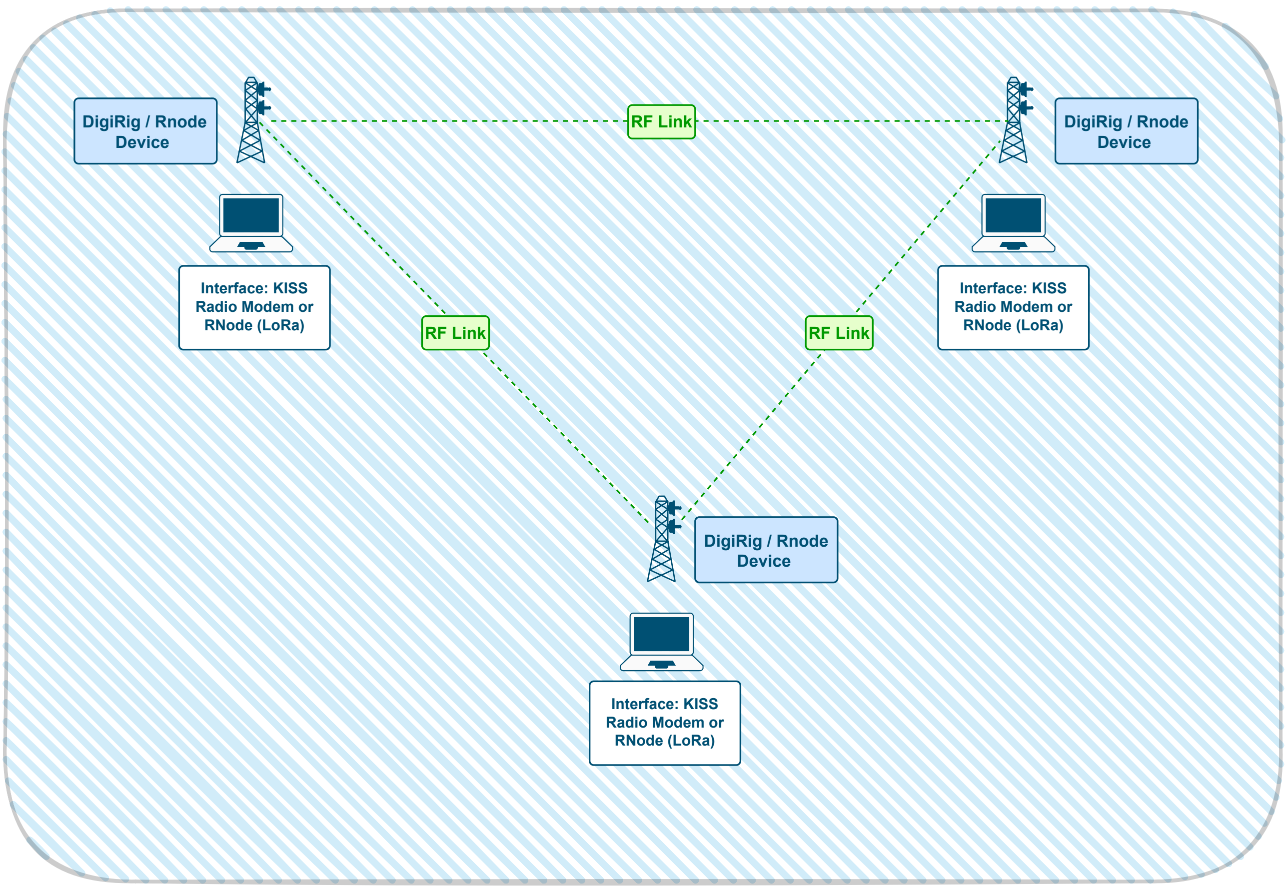
The Phone Book tab is used for sorting prefered stations from the network and is critical to the Group Chat feature. If the station is not in your phone book, and not running Ghostnet, they cannot participate in group chats.

Basic Point-to-Point Network Over RF

A basic PTP network may be establised by combining a Gostnet equiped computer with a DiGiRig, Yaesu SCU-17 or RNode device and any applicabale radio. In testing, the following settings were successful:

DigiRig Mobile: RTS
DigiRig Lite: VOX
Yaesu SCU-17: DTR
T-Beam Supreme: Configured with RNode

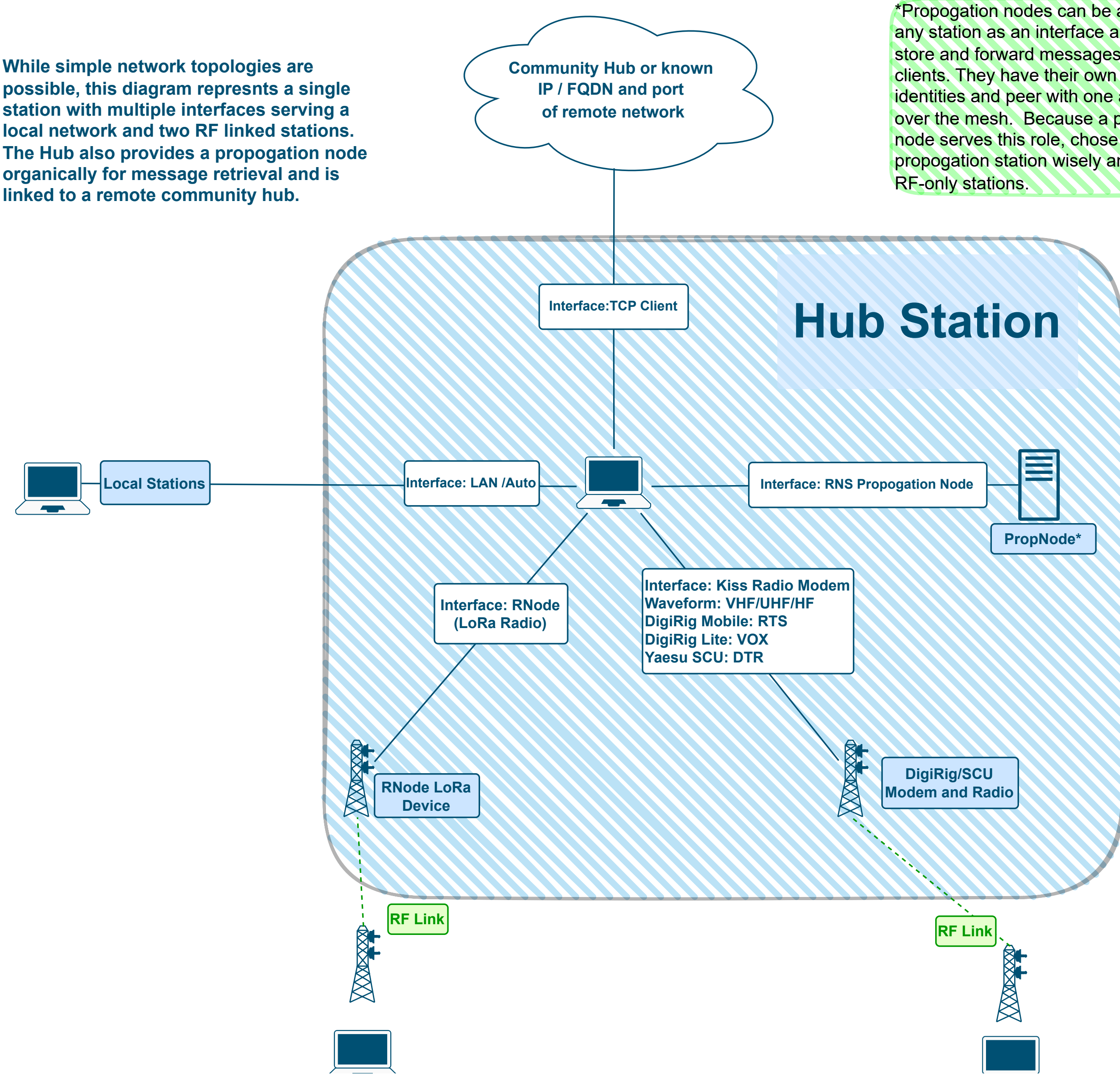
In some parts of the world it is illegal to transmit encrypted data across a radio. Ensure you follow your local laws.

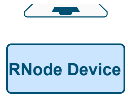


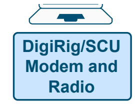
Robust Mesh Network Topology

While simple network topologies are possible, this diagram represnts a single station with multiple interfaces serving a local network and two RF linked stations. The Hub also provides a propogation node organically for message retrieval and is linked to a remote community hub.

*Propogation nodes can be added to any station as an interface and serve to store and forward messages for offline clients. They have their own RNS identities and peer with one another over the mesh. Because a propogation node serves this role, chose the propogation station wisely and avoid RF-only stations.







Group Chat

The Group Chat feature requires all members to run Ghostnet software. Standard Reticulum RNS Identities will not be able to participate.

One member's computer functions as a server/admin and all others are members. Its recomended that the admin station is reliable and running a propogation node interface for the best chances of offline message retrieval.

The admin will identify members in the "network" and select the box next to the identities to add them to the "phonebook". Once members are in the phonebook, they will appear as available for group chat entry.

The admin should "announce" the group chat after its created in the "Admin" button of the "Groups" tab.

RF stations will struggle to retrieve offline messages due to the link requirements of message retreival.